# Gabriel **Pedde Ungureanu** | Physics PhD | GPU-Native CFD & HPC Solver Development

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## Profile

Physicist (PhD, 6 papers) and mathematician (110/110 *cum laude*) who writes **GPU-native solvers for transport physics**: a **Lattice Boltzmann solver built from scratch in CUDA** and a **distributed compressible flow solver** in MPI + OpenACC, each validated against analytical or reference solutions. First-principles modelling over empirical tuning.

## Solver Development — CFD, GPU, Distributed Memory

**Lattice Boltzmann CFD Solver** — GitHub
*CUDA, written from scratch* — *2026*

- **Architecture: D2Q9 BGK solver** for 2D Taylor–Green vortex decay, built from first kernels in CUDA.
- **Optimisation:** restructured distributions **AoS** → **SoA** for coalesced access; added **shared-memory tiling** of the collision step.
- **Performance: 1174 GFLOP/s** and **809 GB/s** sustained bandwidth; ∼**123**× (collision) / ∼**66**× (streaming) over an optimised CPU baseline on a 2048×2048 lattice.
- **Validation:** matched the analytical kinetic-energy decay rate $\nu = (\tau - 1/2)/3$; also ported to **OpenACC**.

**Distributed Compressible Flow Solver** — GitHub
*Fortran, MPI + OpenACC/CUDA-Fortran* — *Dec 2025 – Mar 2026*

- **Numerics:** co-authored a **compressible buoyant-flow solver** (thermal plume in a stratified atmosphere): **flux-form finite volume**, Runge–Kutta integration, dimensional splitting, hydrostatic reference state.
- **Parallelism: MPI domain decomposition with halo exchange**, plus **OpenMP** and **OpenACC/CUDA-Fortran** offload; run on **Leonardo** (CINECA).
- **Validation:** cross-checked serial vs. MPI vs. hybrid vs. GPU — CPU paths bitwise identical, GPU agreeing to $\sim \mathbf{10^{-25}}$ in $L_2$.

**Multi-GPU Jacobi & Distributed Linear Algebra** — GitHub
*MPI + OpenACC, MPI + cuBLAS* — *2026*

- **Distributed solver:** 2D heat equation on a 40,000×40,000 grid — **MPI domain decomposition + OpenACC** offload, with halo exchange overlapped against GPU compute.
- **Scaling:** strong-scaling study across pure MPI, non-blocking MPI and hybrid MPI+GPU (to 16 ranks), timing communication, host–device transfer and compute separately.
- **Dense linear algebra: MPI + cuBLAS** matrix multiplication with a Cannon-style block decomposition.

**Numerical PDEs & Solver Validation** — GitHub
*PETSc/SLEPc, Python* — *2026*

- **Cross-validation:** stiff **nonlinear parabolic PDE** solved two independent ways — method-of-lines **finite differences** (adaptive RK45) and a **physics-informed neural network** — with its fixed point located by **shooting**: critical exponent $\nu = 0.649$ vs. 0.630 from Monte Carlo. Elliptic/Poisson and nonlinear eigenvalue solvers in **PETSc/SLEPc**.

## Research & Engineering Experience

**SISSA** — **Trieste, Italy**
*Doctoral Researcher — Theoretical Physics & Scientific Computing* — *2022–2026*

- Exact, non-perturbative results in quantum field theory through **large-scale symbolic and numerical computation** on SISSA's Ulysses EuroHPC cluster; **6 published papers,** ∼**60 citations**, presented at international conferences and seminars.
- **Verification-first development**: every candidate result is refereed by an independent deterministic evaluator rather than self-graded — the discipline I bring to solver validation.

## Education

**SISSA / ICTP Joint Programme** — **Trieste, Italy**
*Master in High Performance Computing and AI (MHPC)* — *2025–2026*

Concurrent with the final PhD year. **GPU programming (CUDA, OpenACC)**, numerical methods for PDEs, parallel computing (MPI/OpenMP), computer architecture, PETSc/SLEPc.

**Università Cattolica del Sacro Cuore** — **Brescia, Italy**
*BSc Mathematics & MSc Physics (Laurea Triennale & Laurea Magistrale)* — *2017–2022*

Real & functional analysis, PDEs, probability; continuum and statistical mechanics. **Best graduate of the year** (MSc Physics); **110/110 *cum laude*** in both degrees — Italy's maximum grade with honours, equivalent to First-Class Honours / a 4.0 GPA.

## Technical Skills

**CFD & numerics**: **Lattice Boltzmann (D2Q9, BGK)**, finite volume / finite difference, advection–diffusion, compressible & buoyant flow, Runge–Kutta, stiff PDEs, iterative & eigen solvers, validation against analytical solutions
**GPU**: **CUDA** (kernels, shared memory, constant memory, coalescing, occupancy), OpenACC, cuBLAS, Nsight profiling; GPU memory hierarchy & roofline analysis
**Parallel / HPC**: **MPI** (domain decomposition, halo exchange, blocking & non-blocking, multi-GPU), OpenMP, hybrid MPI+X, SLURM, strong/weak scaling; Leonardo (CINECA), Ulysses (SISSA), LUMI
**Programming**: **C, C++, CUDA, Fortran** (modern, OO); Python (NumPy/SciPy), PETSc/SLEPc, HDF5/NetCDF, CMake, Git, Docker/Apptainer, Linux
**Publications**: INSPIRE-HEP | Google Scholar — *Languages:* Italian (native) | English (C1) | Romanian (A2)